R Quest

A guide for teachers, Sunday-school leaders and parents

Free coding lessons in R for children aged about 7 to 14, in English and Yorùbá.

Free **No sign-up needed** **Works on phones**

exelegch.github.io/rquest · Version 2026-10-17

1. What R Quest is

R Quest teaches children to write real code in R, the language scientists and data analysts use to count, compare and draw charts. Children type short programs, see the answer straight away, and earn stars and badges. Everything runs in the web browser: nothing to install, nothing to pay.

Why R for children?

- One line of R already does something visible: `barplot(c(3, 5, 2))` draws a chart.
- It builds data skills: counting, averages, sorting and charts.
- It connects to school maths and to real life, like market prices, savings and the weather.

What children practise

- Reading and following instructions carefully
- Predicting what code will do, then checking
- Finding and fixing mistakes ("debugging") calmly
- Explaining their thinking to someone else

Faith and language

R Quest is gently Christian: the mascot is a dove, a Bible verse greets the children each day, and some activities use Bible stories (for example, the twelve baskets in John 6). Everything can be switched between English and Yorùbá with one button.

The parts of R Quest

Part	What it is
📚 **Learning path**	8 units, from "Talk to R" to "Young data scientist". Each lesson has three steps (see below). Units also contain exercises, a "Fix the bug" puzzle and a project.
🐞 **Fix the bug**	Broken code with a clue. Children read R's message and repair it. This builds patience and careful reading.
🏗️ **Projects**	One per unit. A checklist of 4–5 parts that ticks itself as children complete them. Projects are optional on the path but needed for the unit certificate.
🎯 **Challenges**	15 puzzles at easy, medium and hard levels. R checks the answer.
🎮 **Games**	20 games written in R (guessing games, quizzes, art and music) with difficulty levels.
🧩 **Today's puzzle**	A new short puzzle every day, with a "puzzle streak" for coming back.
⚡ **Speed quiz**	60 seconds of "What will R print?" questions from lessons the child has done.
🎨 **Free play**	A blank page, a chart maker, an art studio and a music maker.
🏅 **Certificates**	A printable certificate for every finished unit, with the child's name and the skills learned.

2. How a lesson works

Every lesson has the same three steps. Children cannot skip ahead, but they can always go back.

1 · Try it
Run the example code and see what happens. Then change something (a number, a word, a colour) and run it again.

2 · Predict
A short "What will R print?" question. Children must think before they press a button. Right first time earns 2 bonus stars.

3 · Your turn
A small mission that uses the new idea in a new way. R checks it. A hint is there if they are stuck.

Tip: the Predict step is the best moment to talk. Ask "Why do you think that?" before the child answers. Wrong guesses are useful: the explanation appears straight away.

When a child is stuck

1. Read R's message together. R Quest adds a plain-English (or Yorùbá) explanation under every error.
2. Look for the usual suspects: a missing bracket `)`, a missing quote mark `"`, a missing comma, or a capital letter where R wants a small one.
3. Change one small thing at a time, then run again.
4. Press 💡 Hint. Using hints is fine. Ask the child to explain the hint back to you.

3. Getting started

What you need

- Any up-to-date browser (Chrome, Edge, Firefox or Safari) on a computer, tablet or phone.
- Internet for the first visit. R downloads once (about 20 MB) and is then kept on the device.
- No accounts. Each child types a first name or nickname.

In a class or group

- Two children per device works well: one types, one reads the task, then swap.
- On shared devices, use "Edit" on the My stuff page to change the name, or use different browsers.
- Parents can make an optional family account to keep progress on several devices.

Settings that help

On **My stuff → Settings**: bigger text, high contrast, calm mode (less movement), sounds on or off, and language. The 🔈 button reads tasks aloud on devices that support it.

4. The units

Ages are a guide only. Younger children can go far with a grown-up reading alongside them.

Unit 1 · Talk to R

Ages 7+ · 7 activities

Printing, maths and variables.

By the end, children can:

- Make R print words and numbers
- Use R as a calculator
- Keep things in named boxes (variables)

Lessons: 👋 Say hello, 🔢 Super calculator, 📦 Variables
Exercises: 🏋 Minutes in a day, 🏋 My age box
Fix the bug: 🐞 missing bracket
Project: 🏗 All about me

Unit 2 · Lists and words

Ages 7+ · 9 activities

Vectors (lists), sorting and picking items.

By the end, children can:

- Make lists (vectors) of numbers and words
- Add up, sort and find the biggest
- Pick items with [ ]

Lessons: 📋 Make a list, 💬 Words too!, 🧮 Math on lists, 🔃 Sorting, 👉 Pick from a list
Exercises: 🏋 Five friends, 🏋 Top score
Fix the bug: 🐞 lost comma
Project: 🏗 My test scores

Unit 3 · Draw with R

Ages 8+ · 8 activities

Scatter, bar, pie and line charts.

By the end, children can:

- Draw scatter, bar, pie and line charts
- Choose colours, shapes and titles
- Say what a chart shows

Lessons: 🎨 Draw dots, 📊 Bar chart, 🥧 Pie chart, 📈 Line chart
Exercises: 🏋 My own bar chart, 🏋 Growing bean
Fix the bug: 🐞 mystery colour
Project: 🏗 Family favourites

Unit 4 · Decisions and loops

Ages 9+ · 10 activities

If/else, loops, functions and random numbers.

By the end, children can:

- Make decisions with if and else
- Repeat things with for and while loops
- Write your own functions

Lessons: ✅ TRUE or FALSE, 🤔 If / else, 🔁 Loops, ⏳ While loops, 🛠️ Functions, 🎲 Random stuff
Exercises: 🏋️ Pass or try again, 🏋️ Seven times table
Fix the bug: 🐞 capital trouble
Project: 🏗️ Times table machine

Unit 5 · R in real life

Ages 9+ · 7 activities

Data frames (tables) and real-world maths.

By the end, children can:

- Keep data in tables (data frames)
- Pick columns with $ and rows with [ ]
- Solve real money and Bible maths problems

Lessons: 🧺 Market day, 🐷 Piggy bank, 🧺 Twelve baskets
Exercises: 🏋️ Shopping total, 🏋️ The four Gospels
Fix the bug: 🐞 wrong column
Project: 🏗️ My mini shop

Unit 6 · Words and text

Ages 9+ · 8 activities

Joining, cutting and changing text.

By the end, children can:

- Join words with paste()
- Change, count and cut words
- Turn words backwards

Lessons: 🔗 Glue words, 🔡 Big and small letters, ✂️ Cut words up, 🔁 Backwards words
Exercises: 🏋️ Shout it out, 🏋️ How long is my name?
Fix the bug: 🐞 lost quote marks
Project: 🏗️ My name poster

Unit 7 · Data detective

Ages 10+ · 9 activities

Real datasets, counts, averages, histograms and box plots.

By the end, children can:

- Explore real datasets
- Count, average and summarise
- Draw histograms and box plots

Lessons: 🌼 Meet a dataset, 🧾 Count and average, 📶 Histogram, 📦 Compare groups, 🏅 Sort a table
Exercises: 🏋️ Average sepal, 🏋️ Longest petal
Fix the bug: 🐞 small letter, big problem
Project: 🏗️ Weather detective

Unit 8 · Young data scientist Ages 12+ 10 activities

Functions that return answers, building results, sapply(), new columns, groups and good charts.

By the end, children can:

- Write functions that give back answers
- Build results with loops and sapply()
- Add columns, group data and make a clear chart

Lessons: 🎁 Functions that give back, 🧱 Build a list in a loop, ⚙️ Do it to every item, 🌽 New columns, 🗂️ Group and summarise, 🖼️ Tell a story with a chart
Exercises: 🏋️ Grade machine, 🏋️ Hottest month
Fix the bug: 🐞 missing piece
Project: 🏗️ Rain report

5. A 45-minute session

Time	What to do
5 min · warm-up	An "unplugged" game with no devices (ideas below). Link it to today's idea.
5 min · show	On one screen, run today's first lesson together. Ask "What do you think will happen if we change this number?"
25 min · code	Pairs work through the next lessons on the path. Fast pairs try the unit's bug fix, challenges or a game.
7 min · share	Two or three pairs show a picture or answer. Celebrate mistakes that were fixed.
3 min · close	One sentence each: "Today I learned...". Optional prayer of thanks for the day's verse.

Unplugged warm-ups

Unit	Warm-up
Talk to R	**Robot teacher:** a child is "R" and only does exactly what is said, word for word. "print hello" → they say "hello".
Lists and words	**Human vector:** five children stand in a line holding numbers. Find the biggest, sort the line, and "pick item 3".
Draw with R	**Class poll:** vote on favourite fruit with hands up, then draw the bar chart on the board before drawing it in R.
Decisions and loops	**If-then game:** "If you are wearing blue, stand up, otherwise clap." Then "Repeat 3 times: jump."
R in real life	**Market table:** write a shopping list with prices on the board and add a total column together.
Words and text	**Word surgery:** cut a printed Bible verse into words, count them, and read one word backwards.
Data detective	**Height line:** children line up by height. Who is in the middle (the median)? What is the range?
Young data scientist	**Function machine:** one child is a "machine" that doubles any number given to them. Others guess the rule.

6. Bible connections

These activities use Bible stories or verses. They are a natural starting point for a short discussion.

Activity	Where	Starting point
🧺 Twelve baskets	Lesson L21	Jesus fed 5,000 people with 5 loaves and 2 fish, and 12 baskets were left over (John 6). Let R do the maths!
🏋 The four Gospels	Exercise E10	Make a list called gospels with the four Gospels: Matthew, Mark, Luke and John. Then print it.
🏋 Shout it out	Exercise E11	Make R print JESUS LOVES ME in capital letters, using toupper().
🌍 Days of creation	Challenge C9	God made the world in six days and rested on the seventh (Genesis 2:2). Add the seventh day to the list.
📖 A verse for you	Game G6	Type your name and R picks a Bible verse for you. Press Play again for another one.
🧠 Memory verse	Game G9	R hides some words so you can practise your memory verse. Harder levels hide more words. You can also type your own verse in the code!
❤️ Heart shape	Game G13	The Bible says God is love (1 John 4:8). This maths formula draws a heart.
✨ Starry night	Game G14	God told Abraham to look at the sky and count the stars, if he could (Genesis 15:5). How many can you draw?
🌈 Noah's rainbow	Game G15	The rainbow is God's promise to Noah (Genesis 9:13). A loop draws each colour, one band at a time.
📚 Bible book finder	Game G17	Type the name of a book of the Bible. R finds where it is and which Testament it belongs to.
🚢 Two by two	Game G18	The animals went into Noah's ark two by two (Genesis 7:9). rep() with each = 2 makes the pairs!
❓ Bible quiz	Game G19	Press Play to see a question. Type your answer in English and press Play again. Pick a harder level when you are ready!
🎵 Music maker	Game G20	R can write music! notes holds the tune and beats says how long each note lasts. This is the start of "Jesus Loves Me".
🌅 Six days and a rest	Today's puzzle D9	God worked six days and rested on the seventh (Genesis 2:2). Use rep() to make a list with "work" 6 times, then add "rest".

Several lesson missions also use Bible words, for example "Thank you, God!" (loops), "God bless you" (functions), Bible books in ABC order (sorting) and Psalm 23:1 (cutting words).

7. Checking progress

- **Parents page** (behind a small maths question): stars, streak, lessons, projects, challenges and games done, stars earned each day this week, minutes of coding, and which skills are complete.
- **Projects** show what a child can do on their own: each part is ticked by R Quest.

- **Certificates** can be printed for each finished unit. Sign them and hand them out. Children love this.

Links to computing standards

R Quest supports these CSTA K–12 Computer Science Standards (2017 edition):

Standard	What it asks	Where in R Quest
1B-AP-09	Create programs that use variables to store and modify data.	Units 1–2
1B-AP-10	Create programs that include sequences, events, loops, and conditionals.	Unit 4, games
1B-AP-15	Test and debug a program or algorithm to ensure it runs as intended.	Every "Fix the bug" puzzle
1B-DA-06	Organize and present collected data visually to highlight relationships and support a claim.	Unit 3, project "Family favourites"
1B-DA-07	Use data to highlight or propose cause-and-effect relationships, predict outcomes, or communicate an idea.	Units 5 and 7
2-AP-11	Create clearly named variables that represent different data types and perform operations on their values.	Units 1, 2 and 6
2-AP-12	Design and iteratively develop programs that combine control structures.	Units 4 and 8
2-AP-14	Create procedures with parameters to organize code and make it easier to reuse.	Functions in units 4 and 8
2-DA-08	Collect data using computational tools and transform the data to make it more useful and reliable.	Units 7 and 8

8. Safety and privacy

- No adverts, no tracking cookies, and data is never sold or shared.
- Without an account, progress stays on the device. Children never need an email address.
- R Quest counts anonymous usage (which lessons are finished or hard) with no names. Families can switch this off on the Parents page.
- Families can delete all their data themselves on the Parents page under **Your data**.
- Please remind children not to type full names, addresses or phone numbers into the 💬 feedback box.

The full privacy notice, in English and Yorùbá, is at **exelegch.github.io/rquest/privacy.html**.

9. Questions

Is it really free? Yes, and it will stay free.

Does it work without the internet? The app opens offline after the first visit, but R itself needs a connection to start. If the connection drops, R starts again by itself when you are back online.

Can I print and adapt the lessons? Yes, for non-commercial teaching. The lessons are under the Creative Commons BY-NC-SA 4.0 licence: share and adapt them for free, name R Quest, and share your changes the same way.

Found a mistake, or have an idea? Use the 💬 button in R Quest, or open an issue at github.com/Exelegch/rquest. Corrections to the Yorùbá are especially welcome.

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